

1.1.

Since actions are chosen uniformly:

Prob. of choosing any action a is:

$$\pi(a) = 1/5$$

Exp. rewards for each arm (given uniform distbⁿ).

$$E_1 = -\frac{3+2}{2} = -0.5$$

$$E_2 = \frac{4+7}{2} = 5.5$$

$$E_3 = \frac{1+4}{2} = 2.5$$

$$E_4 = -\frac{1+9}{2} = 4$$

$$E_5 = \frac{2+5}{2} = 3.5$$

Expected reward:

$$E(R) = \sum_{a=1}^5 \pi(a) \cdot E_a$$

$$= \frac{1}{5} (-0.5 + 5.5 + 2.5 + 4 + 3.5)$$

$$= \underline{\underline{3}}$$